

1. The figures shown in the right margin indicate the full marks for each question.
2. Answer Six questions, taking Three from each section including Q.1, Q.2, Q.5 and Q.6.
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

- | | | Marks |
|------------|--|-----------------------------------|
| Q.1 | <p>(a) Define linear differential equation. Determine the order and degree of the following differential equations and give the reason for your answer:</p> <p>(i) $\frac{\partial^3 u}{\partial x^2 \partial y^3} + \frac{\partial^3 u}{\partial x^3} + \frac{\partial^2 u}{\partial y^2} = 0$, (ii) $\sqrt[4]{\frac{ds}{dt}} = \sqrt{\frac{d^2 s}{dt^2} + 3s - 5}$,</p> <p>(iii) $\frac{d^4 x}{dt^4} + \frac{d^2 x}{dt^2} + \left(\frac{dx}{dt}\right)^5 = e'$. [CLO1, PLO1, C2]</p> <p>(b) Obtain the differential equation of all circles which pass through the origin and whose centers are on the x - axis. [CLO1, PLO1, C3]</p> | <p>1+6</p> <p>3</p> |
| Q.2 | <p>(a) Determine the solution of the following first order ordinary differential equations (any two): [CLO2, PLO1, C3]</p> <p>(i) $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$, (ii) $\frac{dy}{dx} + \frac{x^3+3xy^2}{y^3+3x^2y} = 0$, (iii) $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$.</p> <p>(b) An urban planner wants to predict the population of a city for his planning purpose. He/she consider that the population of the community is increased at a rate proportional to the number of people present at a time t. If the population has doubled in 7 years, how long will it take to triple? [CLO3, PLO2, C3]</p> | <p>6</p> <p>4</p> |
| Q.3 | <p>Solve the following ordinary differential equations (Any three including (i)): [CLO1, PLO2, C3]</p> <p>(i) $(D^4 - 4D^3 + 8D^2 - 8D + 4)y = 0$,</p> <p>(ii) $(D-1)^3(D+2)y = e^x$,</p> <p>(iii) $(D^2 + D)y = x \cos x$,</p> <p>(iv) $(D^2 - 2D + 1)y = x$.</p> | <p>2+4+4</p> |
| Q.4 | <p>(a) Estimate the solution of the following initial value problem: [CLO3, PLO2, C3]</p> <p>$(D^3 - 6D^2 + 11D - 6)y = 0$; $y(0) = 0$, $y'(0) = 0$, $y''(0) = 2$.</p> <p>(b) Determine the particular solution from the following boundary value problem:</p> <p>$y'' + 16y = 0$; $y(0) = 2$, $y(\pi/8) = -5$. [CLO3, PLO2, C3]</p> | <p>5</p> <p>5</p> |

Section-B

Marks

- Q.5 (a) Define each of the terms with an example "semi-linear and quasi-linear partial differential equation of order one". [CLO1, PLO1, C1] 2
- (b) Form a partial differential equation by eliminating the arbitrary function Φ from $\Phi(x+y+z, x^2+y^2-z^2)=0$. [CLO1, PLO1, C3] 4
- (c) Which method do you prefer to solve the following partial differential equation? Using your chosen method, solve the equation $p+3q=5z+\tan(y-3x)$ where $p=\frac{\partial z}{\partial x}$ and $q=\frac{\partial z}{\partial y}$. [CLO2, PLO1, C3] 4
- Q.6 (a) Write down the Charpit's auxiliary equations. Determine the complete integral of the first order non-linear partial differential equation $z^2(p^2z^2+q^2)=1$. [CLO2, PLO2, C3] 5
- (b) Assess the complete solution of the partial differential equation $(D^3-6D^2D'+11DD'^2-6D'^3)z=e^{5x+6y}$. [CLO2, PLO2, C3] 5
- Q.7 (a) What do you mean by central tendency? Discuss different types of central tendency. [CLO1, PLO1, C2] 3
- (b) Calculate the mean, median and standard deviation from the following frequency distribution: [CLO3, PLO2, C3] 7
- | | | | | | | | | |
|----------------|-----|------|-------|-------|-------|-------|-------|-------|
| Class interval | 0-5 | 5-10 | 10-15 | 15-20 | 20-25 | 25-30 | 30-35 | 35-40 |
| Frequency | 2 | 5 | 7 | 13 | 21 | 16 | 8 | 3 |
- Q.8 (a) Define expected value and variance of a random variable. Suppose a fair coin tossed three times. Let X denote the number of heads obtained. Now determine the expected value and variance of X . [CLO3, PLO2, C3] 4
- (b) A hamburger chain found that 75% of all customers use mustard, 80% use ketchup, and 65% use both. What are the probabilities that a ketchup user uses mustard and that a mustard user uses ketchup? [CLO3, PLO2, C3] 3
- (c) Suppose that 15% of visitors to a website are from the USA, 35% are from Australia, and 50% from the rest of the world. The probabilities that a visitor from that region purchases something are 0.01, 0.05 and 0.02 respectively. Now, find the probability that a visitor purchases something. [CLO3, PLO2, C3] 3

Bangladesh Army University of Engineering & Technology (BAUET)

Department of Civil Engineering (CE)

B.Sc. Engineering 1st Year 2nd Semester, Final Examination, Summer-2025

Course Code: CHEM-1201

Full Marks: 60

Course Title: Chemistry-II

Time: 3:00 Hours

N.B.:

1. Figures shown in the right margin indicate full marks
2. Answer Six questions, taking Three from each section including Q. 1, 2, 5, 6
3. Use separate Answer script for each section.
4. Assume reasonable data if necessary.

Section A

	Marks
Q.1 (a) Define white cement and explain its composition. [CO1, PO1, C1, KP1]	2
(b) Compare and contrast the dry process and wet process of cement manufacturing. [CO1, PO1, C2, KP1]	4
(c) The quality of cement is assessed using several tests. Name and briefly describe the various parameters or properties of cement that are commonly tested to ensure its quality. [CO2, PO2, C2, KP2]	4
Q.2 (a) Write the main raw materials used in the manufacture of glass. [CO1, PO1, C1, KP1]	2
(b) Write down the chemical composition of standard (soda-lime) glass. [CO1, PO1, C1, KP1]	4
(c) Explain how color is introduced in glass and discuss the role of different coloring agents and their effects. [CO3, PO2, C3, KP2]	4
Q.3 (a) Define the terms- matrix and reinforcement for composite materials. [CO1, PO1, C1, KP1]	2
(b) Discuss the main factors affecting the quality of composite materials. Explain how each factor influences the mechanical, thermal, or chemical properties of the composite. [CO2, PO2, C2, KP1]	4
(c) Show the key physical, mechanical, and chemical properties of composite materials and describe at least three practical applications of these materials. [CO3, PO2, C3, KP2]	4
Q.4 (a) Define oxidation and reduction in the context of electrochemical reactions. [CO1, PO1, C1, KP1]	2
(b) Differentiate between corrosion and erosion. Highlight the conditions under which each occurs and the materials affected. [CO1, PO1, C2, KP1]	4
(c) Explain wet corrosion and dry corrosion. The study of corrosion is particularly important in Bangladesh- Explain it. Provide examples to support your answer. [CO3, PO2, C3, KP2]	4

Section B

Q.5 (a) Define the terms steel and stainless steel. [CO1, PO1, C1, KP1]	2
(b) Show the major chemical reactions that occurred in a blast furnace during the production of pig iron. [CO3, PO2, C3, KP2]	4
(c) Sketch the figure of a typical blast furnace with appropriate notation. [CO3, PO2, C3, KP2]	4
Q.6 (a) Define the terms isotropy and anisotropy. [CO1, PO1, C1, KP1]	2
(b) Explain the FCC crystal based on atomic radius and atomic packing factor. [CO3, PO2, C3, KP2]	4
(c) Sketch the unit cell of Cr and Zn, and calculate the total number of atoms present per unit cell of each. [CO3, PO2, C3, KP2]	4
Q.7 (a) Define the terms pollutants and contaminants. [CO1, PO1, C1, KP1]	2
(b) Discuss the harmful effects of photochemical smog on the environment. [CO1, PO1, C2, KP1]	4
(c) Show Modified Winkler's method for determining dissolved oxygen (DO) in water. [CO3, PO2, C3, KP2]	4
Q.8 (a) Define the term 'Chromatography'. [CO1, PO1, C1, KP1]	2
(b) Discuss the basic principle of gas chromatography. [CO2, PO1, C2, KP1]	4
(c) Show the Trickling Filter with its merits and demerits. [CO3, PO2, C3, KP2]	4

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3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

- Q.1** (a) Two barrels are being transported in a hand truck as shown in **Figure Q. 1(a)**. **3**
 Draw a free body diagram of each of two barrels. [CLO1, PLO1, C2]

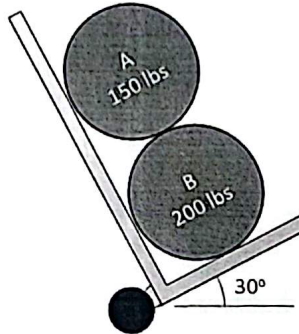


Figure Q. 1(a)

- (b) If the resultant force acting on the bracket is to be 750N directed along the positive X axis, determine the magnitude of F and its direction θ . [CLO1, PLO1, C2] **5**

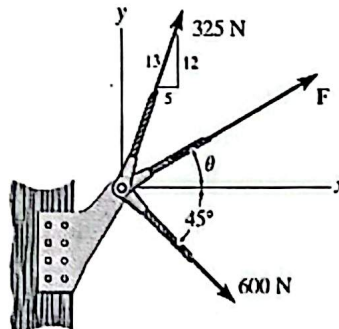


Figure Q. 1(b)

- (c) A continuous string ABCDE, shown in **Figure Q. 1(c)**, passes over smooth pegs at B and D, 10" on centers. To the ends of the strings weights $W_A = 7$ lb and $W_E = 5$ lb are attached. A 10 lb weight is attached at C and the three bodies are in equilibrium. Determine the distance a and the angle α . [CLO1, PLO1, C2] **4**

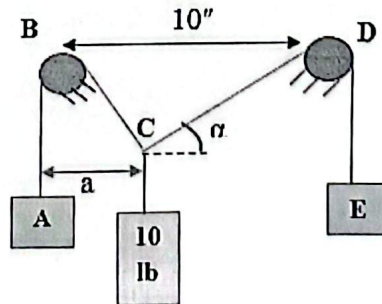


Figure Q. 1(c)

- Q.2 (a) Calculate the support reactions for the beam shown in Figure Q. 2(a).
[CLO2, PLO2, C2]

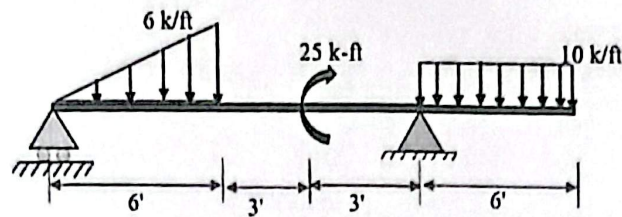


Figure Q. 2(a)

- (b) Calculate the support reactions for the structure shown in Figure Q. 2(b).
[CLO2, PLO2, C2]

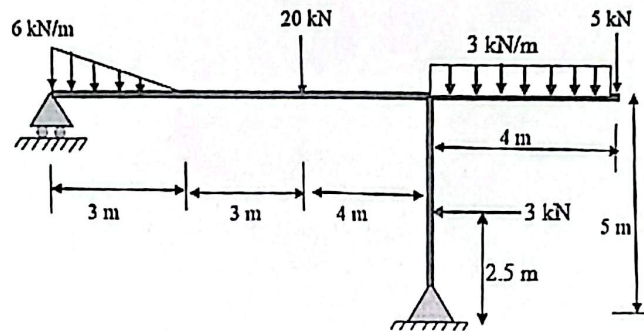


Figure Q. 2(b).

- Q.3 (a) For a parabolic chord, prove that $y = \frac{wx^2}{2Q}$; where symbols bear their usual meanings. [CLO1, PLO1, C2]
- (b) A cable is suspended between two points on the same level with a span of 450 ft and sag of 50 ft. If the load is 2 lb/ft, uniformly distributed horizontally find- (i) the tension in the cable at lowest point (ii) the maximum tension (iii) the slope of the cable at the supports. [CLO1, PLO1, C2]

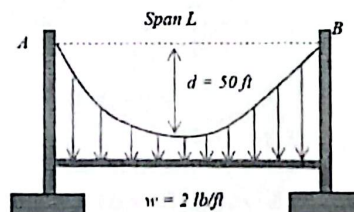


Figure Q. 3(b).

- Q.4 (a) Define coefficient of friction. Distinguish between static and dynamic friction.
[CLO1, PLO1, C2]
- (b) A Particle whose acceleration is $a = 3t - 12 \text{ fps}^2$, is moving at a certain instant in a straight line with an initial velocity of 15 fps in the same sense as the initial acceleration. (a) At the end of 3 sec, what is the velocity of the particle? (b) At the end of 14 sec what is the displacement from origin? [CLO1, PLO1, C2]
- (c) A body resting on a rough horizontal plane, required a pull of 200 N inclined at 30° to the plane just to move it. It was found that a push of 240 N force is required inclined at 30° to the plane just to move the body. Determine the weight of the body and coefficient of friction. [CLO1, PLO1, C2]

Section-B

- Q.5 (a) Compute the centroid of the composite area as shown in **Figure: Q.5 (a)**. [CLO3, PLO2, C3]

6

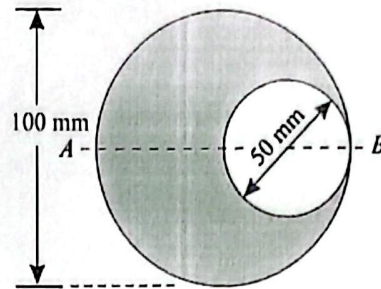


Figure: Q.5 (a)

- (b) Locate the centroid of the shaded area as shown in **Figure: Q.5 (b)**. [CLO3, PLO2, C3]

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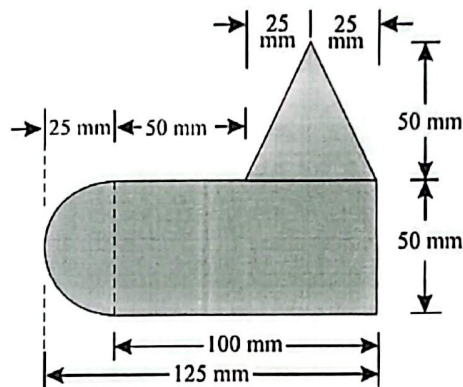


Figure: Q.5 (b)

- Q.6 (a) Calculate the moment of inertia of the section as shown in **Figure: Q.6 (a)** about the axes passing through its center of gravity. [CLO3, PLO2, C3]

6

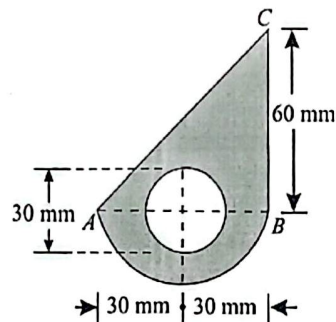


Figure: Q.6 (a)

- (b) For the area as shown in **Figure: Q.6 (b)**, determine the moment of inertia about the axes passing through the center of gravity. [CLO3, PLO2, C3]

6

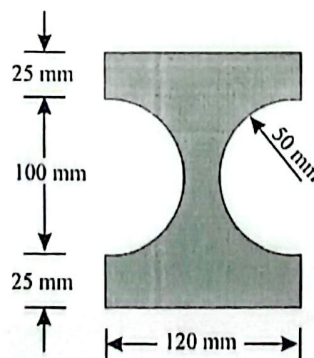


Figure: Q.6 (b)

Q.7 (a)

Two-disc A and B weights 200N and 300N respectively approach each other with velocities of 4 m/s and 2 m/s respectively. If the co-efficient of restitution is 0.6.

[CLO2, PLO2, C3]

Find,

- (i) Their velocities just after impact.
 - (ii) Loss of energy during impact.
 - (iii) If the impact lasts for 0.02 sec, determine the average impact force.
- (b) Three ball A, B, and C of weight 20 N, 40 N, and 60 N are moving in the same direction with velocities 10 m/s, 4 m/s, and 2 m/s respectively. If the ball A strikes with the ball B which in turns strikes with the ball C and the co-efficient of restitution is 0.8. [CLO2, PLO2, C3]

Find,

- (i) Velocities of the balls after impact.
- (ii) Loss of kinetic energy during impact.

Q.8 (a)

Calculate the forces of member of the truss shown in Figure: Q.8(a). Use joint method. [CLO3, PLO2, C3]

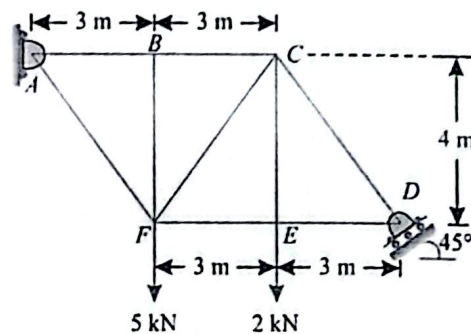


Figure: Q.8(a)

- (b) Determine the nature and magnitude of forces in the members BC, GC, and GF of the truss shown in Figure: Q.8(b). [CLO3, PLO2, C3]

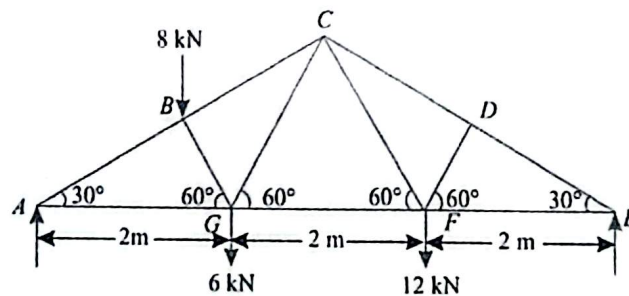


Figure: Q.8(b)

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Section-A

- Q.1** (a) State Ohm's law and define linear resistor. [CO1, PO1, C2, K1] 3
- (b) The charge flowing in a wire is plotted in Fig. 1(b). Sketch the corresponding current. 4
[CO2, PO2, C3, K3]

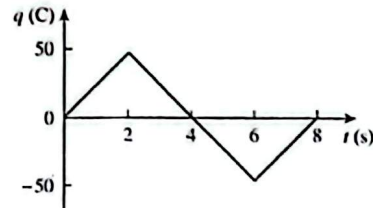


Fig. 1 (b)

- (c) For the following circuit find voltages v_1 and v_2 . [CO2, PO2, C3, K3] 5

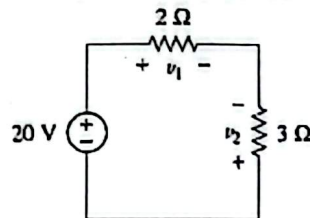


Fig. 1 (c)

- Q.2** (a) Explain the voltage divider rule in a series circuit. [CO1, PO1, C2, K2] 3
- (b) Determine V_o & i in the following circuit shown in Fig. 2 (b). [CO2, PO2, C3, K3] 4

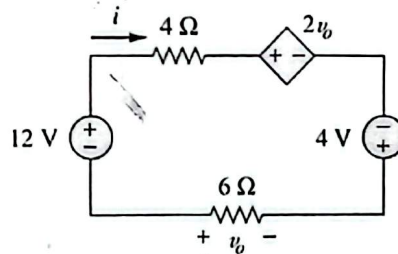


Fig. 2(b)

- (c) Determine v_o & i_o in the following circuit shown in Fig. 2 (c). [CO2, PO2, C3, K3] 5

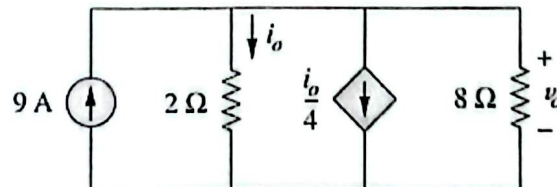


Fig. 2 (c)

- Q.3 (a) Explain the Superposition theorem.
 (b) Determine the voltages at the nodes in the following circuit.

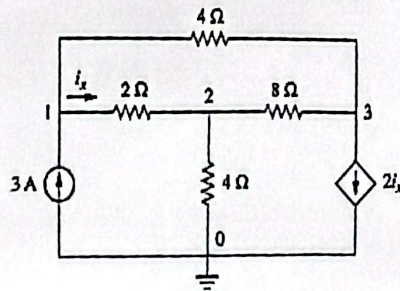


Fig. 3 (b)

- (c) Determine i_1 & i_2 using mesh analysis for the circuit shown in Fig. 3(c).

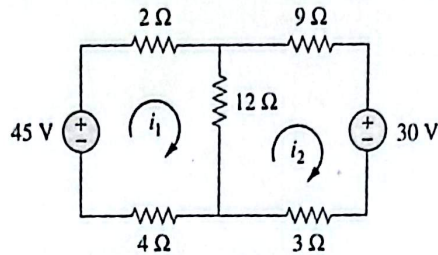


Fig. 3 (c)

- Q.4 (a) Find the amplitude, phase, period, and frequency of the sinusoid: $v(t) = 12\cos(50t + 10^\circ)$.
 (b) Using the superposition theorem, find v_o in the circuit of Fig. 4(b).

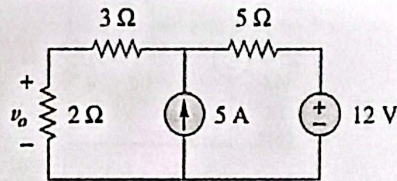


Fig. 4(b)

- (c) Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit of Fig. 4 (c). Then find I .

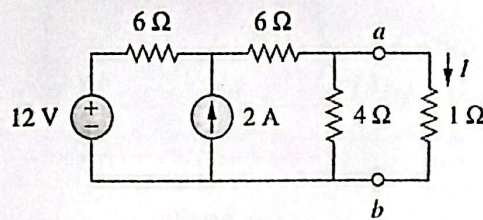


Fig. 4(c)

Section-B

- Q.5 (a) Define phasor. Write down the laws for delta to wye and wye to delta transformation. [CO1, PO1, C2, K1]
 (b) Find the voltage in a circuit described by the integrodifferential equation $2 \frac{dv}{dt} + 5v + 10 \int v dt = 50 \cos(5t - 50^\circ)$ Using phasor approach. [CO2, PO2, C3, K3]

- (c) The current through an element is shown in Fig. 5 (c). Determine the total charge that passed through the element at: (a) $t = 1$ s, (b) $t = 3$ s, (c) $t = 5$ s. [CO2, PO2, C3, K3] 4

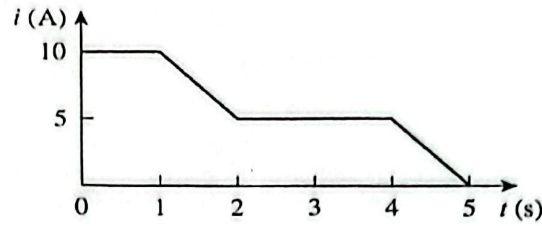


Fig. 5 (c)

- Q.6 (a) What is a rectifier, and why is it used in power supply circuits? [CO2, PO2, C3, K2] 3
 (b) Determine I_D , V_{D2} , and V_o for the circuit of Fig. 6 (b). [CO2, PO2, C2, K3] 4

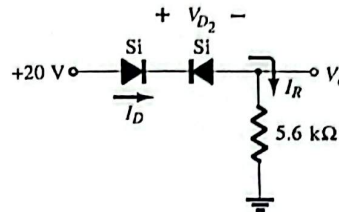


Fig.6(b)

- (c) i. Sketch the output v_o and determine the dc level of the output for the network of Fig. 6 (c). [CO2, PO2, C3, K3] 5
 ii. Repeat part (i) if the ideal diode is replaced by a silicon diode.
 iii. Repeat parts (i) and (ii) if V_m is increased to 200 V, and compare the solutions.

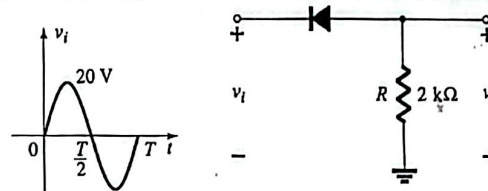


Fig. 6 (c)

- Q.7 (a) State and explain Faraday's Laws of Electromagnetic Induction with examples. 3
 (b) Classify and explain DC generators with appropriate figures. 4
 (c) The following information is given for a 300-kW, 600-V, long-shunt compound generator: Shunt field resistance = 75 Ω , armature resistance including brush resistance = 0.03 Ω , commutating field winding resistance = 0.011 Ω , series field resistance = 0.012 Ω , divertor resistance = 0.036 Ω . When the machine is delivering full load, calculate the voltage and power generated by the armature. 4
- Q.8 (a) Explain the condition for the parallel operation of transformers. 3
 (b) The number of turns on the primary and secondary windings of a single-phase transformer is 350 and 35 respectively. If the primary is connected to a 2.2kV, 50Hz supply, determine the secondary voltage on no-load. 4
 (c) Find the all-day efficiency of 500-kVA distribution transformer whose copper loss and iron loss at full load are 4.5 kW and 3.5 kW respectively. During a day of 24 hours, it is loaded as under: 4

No. of hours	Loading in kW	Power factor
6	400	0.8
10	300	0.75
4	100	0.8
4	0	—